

ESTIMATOR-UPGRADE knob closure: dI-quadrature + amplitude-grid envelopes, a curvature-chord continuum no-condensate bound, and the controlled-error two-shell (0,0) Hessian

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The enumerated single-shell selection $\Delta F[\mathcal{R}] = F[\mathcal{R}] - F[\mathcal{R}_H] > 0$ ($\mathcal{R} \in \{\text{LAM}, \text{HEX}, \text{FCC}, \text{BCC}\}$ at $\mu^2 = 0.005$) was upgraded to a CONTROLLED-ERROR bound against the dominant M -quadrature knob in estimator-upgrade-enumerated v1.0 (curvature $\kappa_{\mathcal{R}} = \Delta F''_{\mathcal{R}}(0) \geq 0.85$, no condensate at $N_{\text{pt}} = 6000$ and 20000). Two numerical knobs and the two-shell ensemble remained. This note closes the two residual single-shell knobs, upgrades the no-condensate verdict from a grid statement to a continuum one, and certifies the two-shell (0,0) Hessian. Every free-energy value is produced by the certified m424 primitives (`dF_reading`, `dI`, `gap_solve`, `M_fast`); the knobs are exercised by monkeypatching m424 module globals, never by re-transcribing the free-energy algebra (code-discipline rule 1).

2. The two residual single-shell knobs and the two-shell question

The free-energy difference uses two quadratures beyond M : the loop log-difference `dI`($\hat{r}, r_R$) (a trapezoid on the grid `_QG` from `_grid_q`($N_{\text{pt}}, q_{\text{max}}$)) and the amplitude scan that decides $\min_{A>0} \Delta F$. The two-shell question is whether the $\{110\} + \{200\}$ ensemble of Math432 – in which the cross moment $m_{31} = \langle \phi_1^3 \phi_2 \rangle \neq 0$ makes the single-shell BCC point non-stationary – preserves the selection.

3. Method and results

(a) amplitude-grid + continuum bound. For each reading the verdict $\min_{A>0} \Delta F > 0$ holds at $N_G = 121/241/481$ and ΔF is strictly increasing on the valid prefix (the unique minimum is $A = 0 = \mathcal{R}_H$). The "closest approach" $\min_{A>0} \Delta F$ itself scales as $\frac{1}{2}\kappa A_{\text{floor}}^2 \rightarrow 0$ as the grid resolves near $A = 0$, so the grid-invariant is the binary verdict, not its magnitude. The node-free guarantee is the curvature-chord bound: a C^2 function deviates below its chord on $[A_i, A_{i+1}]$ by at most $\frac{1}{8}M_i \delta^2$ with M_i the local $|\Delta F''|$ from second differences, so $\Delta F(A) \geq \min(v_i, v_{i+1}) - \frac{1}{8}M_i \delta^2$.

reading	$\min_{A>0} \Delta F$ ($N_G=481$)	monotone	M_{max}	continuum lower bound
LAM	1.17×10^{-6}	yes	126.0	$+8.80 \times 10^{-7}$
HEX	5.84×10^{-7}	yes	358.3	$+4.38 \times 10^{-7}$
FCC	4.76×10^{-7}	yes	526.8	$+3.57 \times 10^{-7}$
BCC	3.38×10^{-7}	yes	629.0	$+2.54 \times 10^{-7}$

The continuum lower bound is positive for every reading: no condensate can hide between grid nodes. Both the value ($\sim \frac{1}{2}\kappa A^2$) and the dip ($\sim \frac{1}{8}\kappa\delta^2$) scale as δ^2 near $A = 0$, and $\frac{1}{2} > \frac{1}{8}$, so the bound survives the near-origin region.

(b) dI-quadrature knob. Refining $_QG$ from $(6000, 50 q_0)$ to $(12000, 100 q_0)$ moves $\kappa_{\mathcal{R}}$ by $< 0.1\%$ and leaves every no-condensate verdict intact.

reading	κ (6000)	κ (12000)	$ \Delta\kappa $
LAM	0.850966	0.850970	4.2×10^{-6}
HEX	2.554165	2.554177	1.3×10^{-5}
FCC	3.407767	3.407784	1.7×10^{-5}
BCC	5.116247	5.116273	2.5×10^{-5}

Combined with the M -knob envelope of estimator-upgrade-enumerated v1.0, the single-shell margins are controlled with respect to all three numerical knobs (M , **dI**, amplitude grid).

(c) two-shell (0,0) Hessian. For the $\{110\} + \{200\}$ condensate (A_1, A_2) the quadratic cross term vanishes: the shells are disjoint, so $\langle\phi_1\phi_2\rangle = 0$ and the m_{31} coupling $U\langle\phi_1^3\phi_2\rangle$ is QUARTIC ($A_1^3A_2$), absent from the Hessian. The Hessian is therefore diagonal, $\kappa_{12} = 0$, with both eigenvalues positive. *Correction (twoshell-continuum-bound v1.0, same change set):* at the B1 point $r_{\text{bare}} = 0.219$ the eigenvalues are $\kappa_{\{110\}} = 5.16$ and $\kappa_{\{200\}} = 3.86$; the SOFTER direction is $\{200\}$, not $\{110\}$ – despite its kernel penalty $Cq_0^4 = 0.214 > 0$, $\{200\}$ has fewer modes ($n_2 = 3 < n_1 = 6$) and the dressing nets a lower curvature. Both positive, so $(0,0)$ is a strict two-shell minimum. The GLOBAL two-shell no-condensate is established as a diagonal-continuum bound at the B1 point in twoshell-continuum-bound v1.0; the earlier Math432 evidence runs at a different operating point $r = 0.005$.

4. What this advances and what remains

ADVANCED: the single-shell enumerated $\Delta F > 0$ verdict is now controlled-error against all three numerical knobs and the no-condensate statement is a continuum (node-free) one; the two-shell $(0,0)$ Hessian is positive-definite with controlled error. REMAINING for full ESTIMATOR-UPGRADE closure: see the dedicated two-shell notes – twoshell-continuum-bound v1.0 (diagonal continuum bound at the B1 point $r = 0.219$) and twoshell-anchored-bracket v1.0 (the EXACT-Wick anchored no-condensate at $r = 0.219$, grid + near-origin continuum). The remaining refinement is a curvature-chord continuum bound on the exact anchored BULK surface. The gate stays OPEN pending that and operator sign-off.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) “The curvature-chord M_i is a discrete estimate of $|\Delta F''|$, not a proven upper bound.” VALID-with-mitigation: M_i is the local second-difference, an $O(\delta^2)$ -accurate estimate; the bound is therefore controlled-error / STRONG-EVIDENCE grade, not a T7 theorem. Mitigation: M_i is grid-stable (the second differences converge as $\delta \rightarrow 0$) and the bound clears zero by a factor ~ 3 near the origin, so a modest M under-estimate cannot flip the sign. Recorded as STRONG-EVIDENCE, not as a proof.

(β) “The two-shell exact-Wick bulk is still grid-grade.” UPHELD: correct. The bracket residual is evaluated directly at the B1 point (twoshell-anchored-bracket v1.0) and does not overturn

positivity, but the exact anchored bulk remains grid-grade until a curvature-chord continuum refinement is run. The two-shell (0,0) Hessian and the diagonal continuum bound are settled content; only the bulk anchored continuum refinement remains (named in section 4).

- (γ) “Which shell is the soft two-shell direction?” The $A_2 = 0$ slice is the single-shell BCC free energy, so the $\{110\}$ curvature at (0,0) is $\kappa_{\text{BCC}} \approx 5.16$. CORRECTED (twoshell-continuum-bound v1.0): the $\{200\}$ direction is SOFTER, $\kappa_{\{200\}} = 3.86 < 5.16$, so the binding (0,0) eigenvalue is 3.86, not 5.116. Both positive; PD stands.

Quantitative sanity check (mandatory): magnitude – κ envelopes $\sim 10^{-5}$ vs $\kappa \sim 1-5$ ($< 0.1\%$); the continuum dip $\frac{1}{8}M\delta^2 \sim 10^{-7}$ vs value $\sim 10^{-6}$; sign-direction – every $\kappa > 0$ and every continuum bound > 0 (the selection-preserving direction); limit-case – $\Delta F(0) = 0$ exactly; consistency – the $\{200\}$ penalty $Cq_0^4 = 0.214 > 0$ does NOT order the curvatures: the binding (0,0) eigenvalue is $\kappa_{\{200\}} = 3.86$ (twoshell-continuum-bound v1.0).

6. Result footer

Result ID:	B1-RH-ENUM / ESTIMATOR-UPGRADE knob closure (single-shell)
Precise statement:	For LAM/HEX/FCC/BCC at $\mu^2=0.005$: (ii) κ_R moves $< 0.1\%$ under dI-grid refinement (6000,50) $\rightarrow$ (12000,100) and the no-condensate verdict is grid-monotone at NG=121/241/481; (iii) the curvature-chord continuum lower bound $\min(v_i, v_{i+1}) - (1/8) M_i \delta^2 > 0$ on every $A>0$ interval (no inter-node condensate); (i) the two-shell $\{110\}+\{200\}$ (0,0) Hessian is positive-definite ($\kappa_{12}=0$; both eigenvalues >0). CORRECTED (twoshell-continuum-bound v1.0): soft eigenvalue = $\kappa_{\{200\}}=3.86$ NOT 5.116; global two-shell no-condensate = diagonal-continuum bound at B1 $r=0.219$ (Math432 ran at a different operating point $r=0.005$).
Scope:	Single-shell ensemble, all 3 numerical knobs + continuum no-condensate; two-shell (0,0) curvature only.
Dependencies:	Math424_AddA_reading_uniqueness (dF_reading, dI, gap_solve, M_fast); estimator-upgrade-enumerated v1.0 (M-knob); Math432 (two-shell global no-condensate).
Evidence grade:	EXECUTED (estimator_upgrade_knobs.py 13/13) + STRONG-EVIDENCE (curvature-chord continuum bound).
Reproduction command:	python codes/vacuum/estimator_upgrade_knobs.py
Expected output:	13/13 PASS; continuum lower bounds positive; dI envelope $< 0.1\%$; two-shell Hessian PD.
Falsification gate:	A reading with a negative continuum lower bound; a dI envelope $> 0.1\%$ of κ ; a two-shell direction with non-positive (0,0) curvature; or a Math432 reload with an anchored dF < 0 .
Tier before / after:	No tier/gate flip. ESTIMATOR-UPGRADE: single-shell knob closure + continuum no-condensate + two-shell (0,0) curvature ADVANCE; gate stays OPEN pending the bulk anchored continuum refinement + operator sign-off (the exact-Wick bracket is closed: twoshell-anchored-bracket v1.0).
No-overclaim statement:	Controlled error w.r.t. the named numerical knobs and the two-shell (0,0) curvature only. NOT a full ESTIMATOR-UPGRADE closure: the two-shell global no-condensate is established at $r=0.219$ (twoshell-continuum-bound + twoshell-anchored-bracket v1.0); the Math432 $r=0.005$ citation is superseded. The B1 T6 selection SIGN was already established; this

Next required action: upgrades the enumerated MARGINS' error grade.
Curvature-chord continuum bound on the exact anchored
BULK surface (the only remaining refinement); then
ESTIMATOR-UPGRADE closure is an operator decision.